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COMMENTARY

Towards Environmentally Extended Health Economic Analysis Plans: A Case Study for the MEL-SELF Trial

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1 Introduction

Health economic evaluations compare the costs and consequences of different healthcare interventions [1]. They are often conducted alongside randomised controlled trials (RCTs) to increase translation of trial outcomes into policy, particularly reimbursement decisions.

The conduct of an economic evaluation can be prespecified in a health economic analysis plan (HEAP) [2]. By prespecifying methods intended for the economic evaluation, HEAPs reduce bias from the selective reporting of results. This is similar to the benefits of prespecifying statistical analyses in a trial's statistical analysis plan. A recent Delphi study by Thorn et al. proposed essential items to include in HEAPs, improving consistency across trials. These items relate to: administrative information, trial background, economic approach, economic data collection and management, economic data analysis, modelling and reporting [3].

Health economic evaluations traditionally focus on health system costs and health outcomes, and the content of a HEAP mostly relates to these outcomes. However, there has been recent academic interest in methods to include environmental impacts in economic evaluations [4]. Several examples of this type of analysis have been published in recent literature [5–10]. Environmental impacts can include greenhouse gas emissions that contribute to climate change, other emissions, water or land use, and physical waste. Methods for how these impacts are included and analysed in economic evaluations vary and there are inconsistencies in how these analyses are planned and reported. Health economic analysis plans, like economic evaluations, have not traditionally included details about environmental impacts, and guidance for their content is silent on environmental impacts.

If health economic evaluations of RCTs are to routinely include an analysis of environmental impacts, then guidance for HEAPs must be updated or extended to reflect this. In this article, we outline items included in a HEAP for the economic evaluation of the Melanoma SELF surveillance (MEL-SELF) RCT; detailed descriptions of the trial are available elsewhere [11, 12].

The MEL-SELF RCT is a two-arm, parallel, superiority RCT with 504 participants and a 1:1 allocation ratio between the intervention and control. In the control arm, patients receive 'clinician-led surveillance' (i.e. usual care: routinely scheduled clinic visits for a total body skin examination). In the intervention arm, patients receive 'patient-led surveillance', which includes usual care plus a mobile dermatoscope, a smartphone application for store-and-forward teledermatology, fast-tracked unscheduled clinic visits if recommended by the teledermatologist, instructions and training on how to use the dermatoscope and teledermatology, access to video guidance on how to complete a total skin self-examination and reminders to complete a skin self-examination. Findings from a pilot trial and costing study were reported previously [13, 14]. Details of the planned cost consequence analysis have been pre-specified in a HEAP, which is available on the Open Science Framework (https://doi.org/10.17605/OSF.IO/65XPW). Version 1.0 of the HEAP is included in the Electronic Supplementary Material of this article.

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2 Items for an Environmentally Extended HEAP

Methods to include environmental impacts in economic evaluations vary and can include approaches that report environmental impacts parallel to the traditional analysis of costs and benefits, though to the full integration of environmental impacts into the incremental cost-effectiveness ratio. Details of these methods are available elsewhere [4, 15–19]. In this article, we describe 14 items in the MEL-SELF HEAP relating to the analysis of environmental impacts. The majority of these items ($n = 12$) are based on the those previously proposed by Thorn et al. [3]. Modifications to these items were made to describe the analysis of environmental impact relevant to each item. This was done to suit the specific analysis of the MEL-SELF Trial in consultation with study investigators, based on work ongoing by Oldenburg et al., and with environmental scientists and health economists [20]. The items are detailed in Table 1, with extracts of text from the MEL-SELF HEAP provided for each.

The first three items were contextual and helped to identify the planned economic evaluation as one that will include environmental impacts. This was done by including reference to environmental impacts in the trial background and rationale (Thorn et al. item 2.1), the economic evaluation aims (Thorn et al. item 3.1) and objectives (Thorn et al. item 3.2). Broad ambiguous terms such as "sustainability" were avoided and the specific type of environmental impact was stated (i.e. carbon emissions).

Then, items were included to describe the methods by which environmental impacts will be included in the evaluation. This was included in the overview of the economic analysis (Thorn et al. item 3.3) with reference to the approach taken, as described by Toolan et al. (i.e., parallel approach) [21]. Further details were specified in a new item to describe the environmental impact of resource use, including details about how environmental impacts will be measured and valued. Technical details were reported for the perspective (Thorn et al. item 3.5), time horizon (Thorn et al. item 3.6), discount rate (Thorn et al. item 5.3) and cost-effectiveness threshold (Thorn et al. item 5.4), and how these will be applied to environmental impacts.

Finally, details were provided about the analysis of environmental impacts in a new item. Further details of the analysis are provided in items about missing data (Thorn et al. item 5.10), analysis of cost effectiveness (Thorn et al. item 5.11), subgroup analysis (Thorn et al. item 5.13) and sensitivity analysis (Thorn et al. item 5.15). In the MEL-SELF RCT example, no items were included from 'Section 6: Modelling' because the planned economic evaluation is a within-trial analysis and does not include any modelling.

3 Discussion

This is the first study to include environmental impacts in a HEAP and contributes to ongoing work exploring how environmental impacts can be included in health economic evaluations. The reporting of health economic evaluations, after the analysis is completed, is guided by the Consolidated Health Economics Evaluation Reporting Standard (CHEERS), which helps ensure studies are identifiable, interpretable and useful for decision making [22]. Work is underway to develop an extension to the CHEERS checklist that includes climate impacts (CHEERS-ClimatE) [20]. Including environmental impacts in both HEAPs and the CHEERS reporting guidelines will be an important step towards routinely considering environmental impacts in economic evaluations.

Health economic analysis plans improve transparency and reduce the risk of bias in economic evaluations [3], and by extending the HEAP these benefits will also apply to the analysis of environmental impact. Doing so may help reduce the risk of green washing, where analyses are selectively reported to make an intervention appear more environmentally favourable than it truly is. It may also encourage communication between clinical trialists, statisticians, and health economists about the analysis of environmental impact at an early stage. This will ensure that key methodological decisions are considered in detail, some of which are areas of active research.

The HEAP items described in this study are appropriate for a cost-consequence analysis conducted alongside a RCT, such as in the planned MEL-SELF economic evaluation. The primary outcome of the trial is the proportion of patients diagnosed with a melanoma at an unscheduled clinic visit, which is being used as a proxy for early detection of melanoma. The willingness to pay for an increase in this outcome has not been established, which is why a cost-consequence rather than a cost-effectiveness analysis is planned for this trial. The HEAP items described in this study may not fully capture those required for modelled cost-effectiveness or cost-utility analyses. Future work may consider these study designs and additional details necessary to include in a HEAP describing these types of study. This may include a description of whether environmental impacts will be integrated into the evaluation, and if so how this will be done. Should willingness to pay for a melanoma diagnosed at an unscheduled clinic visit be determined, a cost-effectiveness analysis of the MEL-SELF trial may be performed, and these items considered in more detail.

Future research could use a Delphi process to build a consensus for items to include in an environmentally extended HEAP, consistent with similar work ongoing for the CHEERS reporting guidelines. The items included in

Table 1 Items from the MEL-SELF trial HEAP describing the planned analysis of environmental impacts in the economic evaluation

Thorn et al. [3] Item	Description	Guidance for environmental impact	Example from MEL-SELF HEAP (Version 1)
Section 2: Trial information and background			
2.1	Trial background and rationale	Describe why environmental impacts are relevant and briefly justify their inclusion in the study	Theoretical benefits from clinician-led surveillance from earlier detection and treatment currently lack robust empirical evidence from randomised trials. There is evidence that clinic visits are associated with health system and out-of-pocket costs, opportunity costs, psychological harms, and carbon emissions. [Page 8, Section 2.1]
Section 3: Economic approach/overview			
3.1	Aim(s) of economic evaluation	Reference environmental impacts in the aim of the economic evaluation	The aim of this economic evaluation is to perform a cost-consequence analysis and quantify the health outcomes, costs, and carbon emissions of patient-led surveillance compared with clinician-led surveillance as a mode of follow up for patients treated for an early-stage melanoma. [Page 11, Section 3.1]
3.2	Objective(s) of economic evaluation	Reference environmental impacts in the objectives of the economic evaluation	We will quantify the health outcomes, health system costs, out-of-pocket costs, productivity changes, and carbon emissions of patient-led surveillance compared to clinician-led surveillance over a period of 24-months from the health system and societal perspectives. We will also compare two types of dermatoscope (polarised or non-polarised light source) used by patients receiving patient-led surveillance. [Page 11, Section 3.2]
3.3	Overview of economic analysis	Outline the approach to how environmental impacts will be included in the economic evaluation. This should include basic information about the approach or method used to include environmental impacts in the study	The within-trial economic evaluation will be performed using individual patient-level data from the MEL-SELF trial. These data will be linked to the New South Wales Cancer Registry and the Medicare Benefits Schedule (MBS). Disaggregated resource use, healthcare costs, environmental impacts, and health impacts will be presented as the mean costs per participant over the 24-month study period, by randomised treatment allocation. Mean health impacts will be estimated in consultation with the MEL-SELF trial statistician. Environmental impacts will be reported using a parallel approach and not be valued in health or monetary units or included in any summary ratios. [Page 11, Section 3.3]
3.5	Perspective(s)	Include any reference to environmental impacts, as applicable	The base case analysis will be conducted from the health system perspective. A secondary analysis will be conducted from the societal perspective and include health system costs, patient out-of-pocket costs, productivity changes, and environmental impacts. [Page 11, Section 3.5]
3.6	Time horizon(s)	State the time horizon for the calculation of environmental impacts	The within-trial economic evaluation will be conducted over a period of two years, using data collected from randomisation to the end of the study follow up period. The time horizon of environmental impacts used in the study will be reported in an appendix to the manuscript. [Page 11, Section 3.6]

Table 1 (continued)

Thorn et al. [3] Item	Description	Guidance for environmental impact	Example from MEL-SELF HEAP (Version 1)
Section 4: Economic data collection and management			
New	Environmental impact of resource-use data	Identify which environmental impacts will be included, how they will be measured (units) and (if applicable) how they will be valued	Carbon emissions will be included in kg carbon dioxide equivalent units (CO2-e). Other environmental impacts will not be considered. The carbon emissions of all resources identified in section 5.6 will be included by considering the carbon emission of the resource itself and of patient travel. Emissions from services will be identified from existing process-based life cycle assessments, such as those listed in the HealthcareLCA database (www.healthcarelca.com) or Lancet MedZero database (https://www.medzerocarbon.com). The quality of estimates will be interrogated before using them. Where existing high quality data are not available assumptions will be made and documented based on similar healthcare resources. Emissions from patient transport will be included based on travel distance and mode of transport (e.g., walk, car, train etc) information recorded in the monthly healthcare use diary. Emissions factors for transport will be obtained from the Australian National Greenhouse Accounts Factors and EcoInvent. Environmental impacts will not be valued in monetary units. [Page 14, Section 4.9]
Section 5: Economic data analysis			
5.3	Discount rates	Detail the source of, and justification for, discount rates used for costs, benefits and environmental impacts	A discount rate of 5% will be used for costs and health outcomes in the second year of the study, consistent with recommendations by the Australian Pharmaceutical Benefits Advisory Committee (PBAC) and Medical Services Advisory Committee (MSAC). Carbon emissions (in kg CO2-e) will not be discounted in this study, to ensure intergenerational equity. This is consistent with the approach recommended by McAlister et al. [Page 16, Item 5.3]
5.4	Cost-effectiveness threshold	Describe if the inclusion of environmental impacts will change any cost-effectiveness thresholds used	No cost-effectiveness threshold will be used (as per the cost-consequence study design), and all costs, outcomes, and environmental impacts will be reported in a disaggregated format. [Page 16, Item 5.4]
New	Analysis of environmental impact	Describe analyses of environmental impact data	Mean per-patient carbon emissions and standard deviations will be summarised for each treatment group. Mean difference between the intervention and control arms will be calculated with 95% confidence intervals. [Page 17, Item 5.8]

Table 1 (continued)

Thorn et al. [3] Item	Description	Guidance for environmental impact	Example from MEL-SELF HEAP (Version 1)
5.10	Missing data	Specify the procedure for dealing with missing environmental impact data	Preference will be given to linked Medicare data, and this will be supplemented by information from patient healthcare diaries and clinic records. Missing data is possible due to patient withdrawal, non-response, or loss to follow-up. Descriptive statistics will be used to explore the amount of missing data, patterns in missing data, and the association between missingness, baseline variables, and outcomes. If appropriate, the base case analysis will assume that data is missing at random (MAR). Multiple imputation methods will be used for cost and environmental impact data. Retrieved dropout multiple imputation or mixed model will be used for health outcomes, consistent with the MEL-SELF SAP. Sensitivity analyses will be conducted to explore departures from the MAR assumption (or other assumed methods), and this may include methods such as reference-based multiple imputation. [Page 17, Item 5.11]
5.11	Analysis of cost effectiveness	Describe methods for how cost effectiveness will be summarised. Include how environmental impacts will be included or excluded from these results	The outcomes, costs, and environmental impacts will be reported in disaggregated formats. Tables will present summary data by treatment group. [Page 17, Item 5.12]
5.13	Subgroup analysis	Describe any analyses of subgroups or heterogeneity in cost effectiveness and the analysis methods used	Analysis will also be conducted to investigate how outcomes, costs, carbon emissions, vary between different patient subgroups. Subgroup analyses may include sociodemographic (e.g., age, sex, socioeconomic status, remoteness) and clinical variables (e.g., treatment centre, melanoma substage, dysplastic naevus syndrome, risk of subsequent new primary melanoma). Subgroups may be omitted from the analysis based on their sample size. [Page 18, Section 5.14]
5.14	Sensitivity analysis	Describe any sensitivity analyses and their form related to environmental impacts	A sensitivity analysis will be conducted that applies a 5% p.a. discount to environmental impact. [Page 18, Section 5.15]

Note: Items provided by Thorn et al. are not always equivalent to section numbers in the MEL-SELF HEAP because of the addition of two items

HEAP health economic analysis plan, *SAP* statistical analysis plan

this study could be used to inform this work. Additional validation of the items will also be needed to determine their applicability to different evaluations answering specific economic questions.

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Data availability The Health Economic Analysis Plan described in this study is included in the Electronic Supplementary and is available on the Open Science Framework (https://doi.org/10.17605/OSF.IO/65XPW). No other data was generated by this study.

Declarations

Conflict of interest Scott McAlister has no conflicts of interest that are directly relevant to the content of this article.

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